ENVS 193DS Final

Iris Carnahan

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Link to GitHub repo

https://github.com/iriscarnahan05/ENVS-193DS_spring-2026_final

Set-up tasks

```
library(tidyverse) # general use
library(here) # cleaning data frames
library(janitor) # file/folder organization
library(readxl) # reading .xlsx files
library(ggeffects) # calculating the data
library(dplyr)
library(forcats)
library(MuMIn)
library(DHARMa)
# reading in `nest_data_final` csv file
nest_data <- read.csv(here("data", "nest_data_final.csv"))
bike_data <- read.csv(here("data", "biking_data.csv"))
```

Problem 1

a. Transparent statistical methods

In part 1, they used a **binomial generalized linear model (GLM), or a logistic regression**. In part 2, they used a **chi-squared test**.

b. Suggestions for rewriting

In the San Francisco Bay, the distance from the water edge tends to predict the likelihood of microhabitat use by American Avocets.

Noticeably, certain shorebirds have shown preference for different habitat types in the San Francisco Bay.

c. Further analysis

The amount of restored native plant cover (measured in % cover) could influence the probability of American Avocet microhabitat use, because they tend to prefer wide-open areas rather than highly vegetated ones in order to forage and feed. Measuring in percentage makes this variable continuous, and this data could be collected via quadrat estimates.

Problem 2

a. Clean and wrangle

```
# creating data frame called `nests_clean`
nests_clean <- nest_data |>
  # selecting columns of interest
  select(height_cm, case_control, ant_species) |>
  # filtering to only include ant species AB, RRB, and BBR
  filter(ant_species == "AB" |
         ant_species == "RRB" |
         ant_species == "BBR") |>
  # reordering ant species
  mutate(ant_species = factor(ant_species),
         ant_species = fct_relevel(
         ant_species,
         "AB", "RRB", "BBR")) |>
  # creating new column to store scientific names
  mutate(scientific_name = case_when(
    ant_species == "AB" ~ "C. sjostedti",
    ant_species == "RRB" ~ "C. mimosae",
    ant_species == "BBR" ~ "C. nigriceps",
    # fills in NA to match with ant species NA
    TRUE ~ NA_character_
  ))
#display 10 rows of object
slice_sample(n = 10,
nests_clean)
```

```
   height_cm case_control ant_species scientific_name
1      302.5            0         RRB      C. mimosae
2      181.0            1         RRB      C. mimosae
3      153.5            0         RRB      C. mimosae
```

```
4       171.0              0          RRB       C. mimosae
5       211.5              0          RRB       C. mimosae
6       155.0              0          BBR    C. nigriceps
7       114.0              0          RRB       C. mimosae
8       182.0              0          RRB       C. mimosae
9       368.0              0          RRB       C. mimosae
10      163.5              0          RRB       C. mimosae
```

b. Response variable

The response variable is the occurrence of a bird nest of any species. The 0 indicates that there is no nest, and the 1 indicates that there is a bird's nest.

c. Purpose of study

Tree height is a continuous variable measured in centimeters. The research shows in the second paragraph of the results section that the taller the tree, the more likely it is that a bird will have nested in in it.

Ant species are shown to be a categorical variable, as shown at the end of the first paragraph in the Methods section of the research paper. The third paragraph of the results showed that trees inhabited by *C. mimosae* and *C. nigriceps* were more likely to contain nests.

d. Table of models

Model number	Ant Species	Tree Height	Interaction
0			
1	X	X	
2	X	X	X
3	X		
4		X	

e. Run the models

```
# no interaction
nest_mod1 <- glm(case_control ~ ant_species + height_cm,
             data = nests_clean,
             family = "binomial") # indication of glm test type
```

```r
# interaction
nest_mod2 <- glm(case_control ~ ant_species * height_cm,
                data = nests_clean,
                family = "binomial") # indication of glm test type
```

```r
# model 0: null model
model0 <- glm(
  case_control ~ 1, # formula: no predictors
  data = nests_clean) # clean data frame
# model 1: all predictors
model1 <- glm(
  case_control ~ ant_species + height_cm,
  data = nests_clean)
# model 3: ant species only
# not model 2 because that was already used above
model3 <- glm(
  case_control ~ ant_species,
  data = nests_clean
)
# model 4: tree height only
model4 <- glm(
  case_control ~ height_cm,
  data = nests_clean
)
```

f. Select the best model

```r
# calculating AIC
AICc(
  model0,
  model1,
  model3,
  model4,
  nest_mod2
) |>
  # arranging models in order of AIC
  arrange(AICc)
```

```
          df     AICc
nest_mod2  6 238.1236
```

```
model1      5 263.9785
model4      3 270.7722
model3      4 295.9079
model0      2 296.3160
```

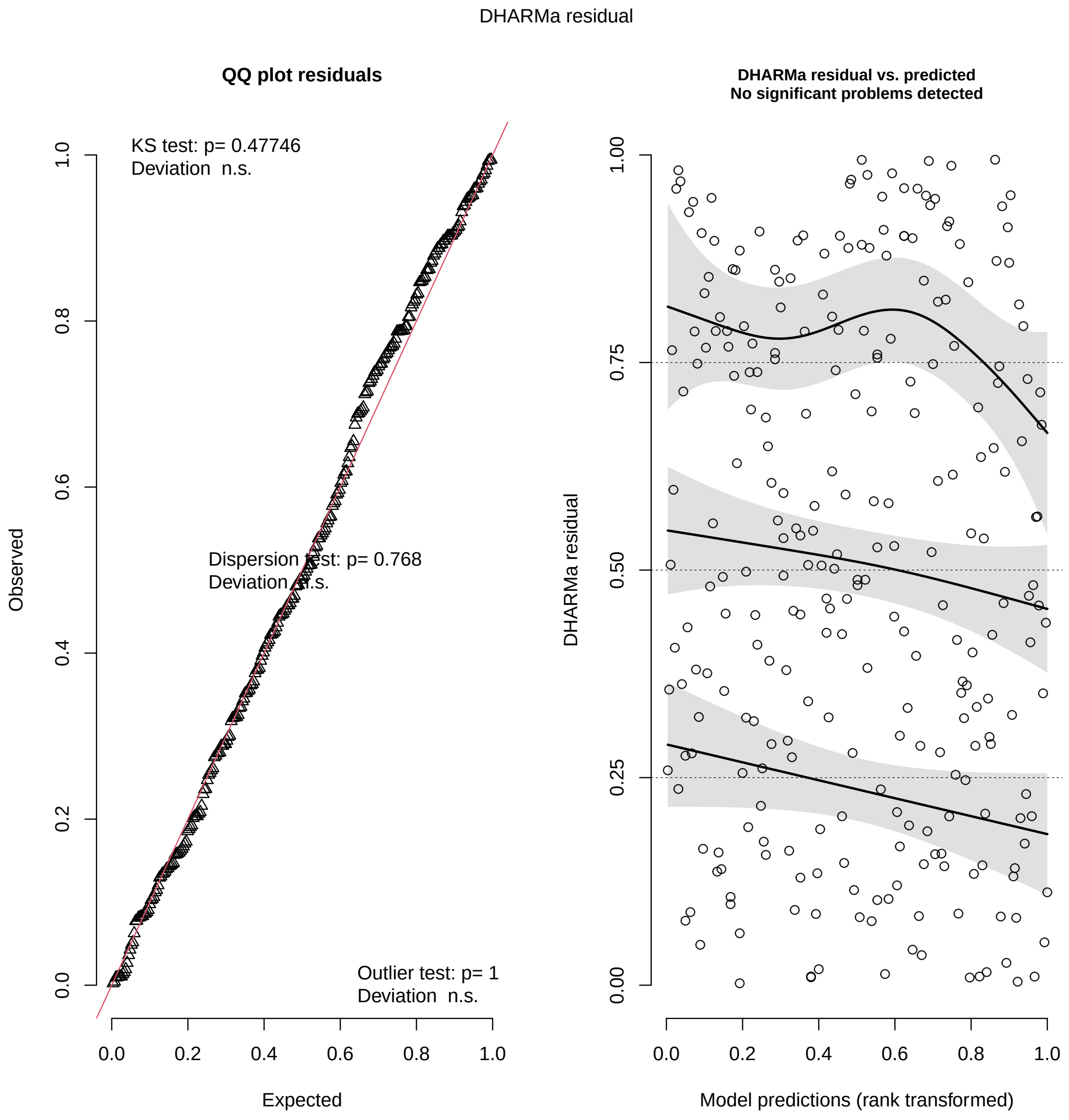
```r
# summarizing model with interaction
summary(nest_mod2)
```

```

Call:
glm(formula = case_control ~ ant_species * height_cm, family = "binomial",
    data = nests_clean)

Coefficients:
                          Estimate Std. Error z value Pr(>|z|)
(Intercept)              -3.693410   2.517395  -1.467  0.14233
ant_speciesRRB            0.848334   2.554617   0.332  0.73983
ant_speciesBBR          -12.279738   5.946723  -2.065  0.03893 *
height_cm                 0.003895   0.008237   0.473  0.63634
ant_speciesRRB:height_cm  0.002597   0.008386   0.310  0.75679
ant_speciesBBR:height_cm  0.084541   0.031961   2.645  0.00817 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 286.04  on 269  degrees of freedom
Residual deviance: 225.80  on 264  degrees of freedom
AIC: 237.8

Number of Fisher Scoring iterations: 7
```

The best model as determined by Akaike's Information Criterion (AIC) is the one where both predictor variables of ant species and tree height (cm) had an interactive effect on the presence of a nest. Accounting for both predictive variables and the potential effects they have on each other best predicts the data outcome.

g. Check the diagnostics

```r
# creating diagnostic plots
plot(simulateResiduals(nest_mod2))
```

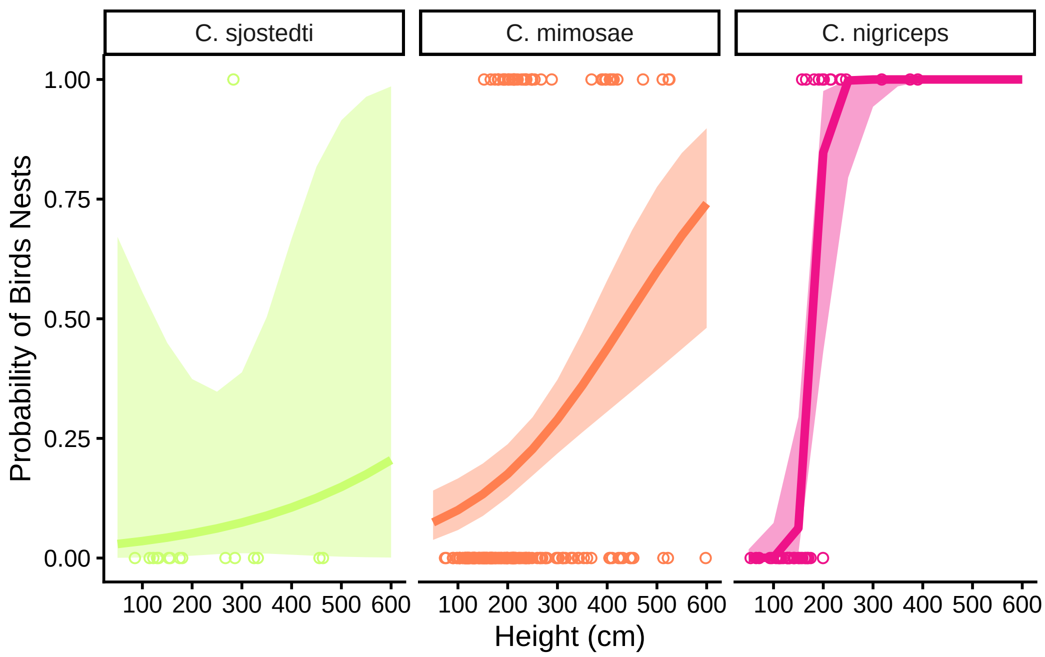


h. Visualize the model predictions

```r
# creating ggpredict data frame
pred <- ggpredict(nest_mod2,
          terms = c("height_cm",
                    "ant_species"))
# renaming `group` as `ant_species` for uniformity
pred <- rename(pred, "ant_species" = "group")
```

```r
# base layer: ggplot
ggplot(data = nests_clean,
       aes(x = height_cm, # x: height of trees
           y = case_control)) + #y: nesting probability
  # first layer: geom point
  # change shape to open circles
  geom_point(aes(color = ant_species),
             shape = 21) +
  # second layer: geom line
  # color based upon ant species
  # adjust line width
  geom_line(data = pred,
            aes(x = x,
                y = predicted,
                color = ant_species),
            linewidth = 1.5) +
  # third layer: geom ribbon
  # change transparency
  geom_ribbon(data = pred,
              aes(x = x,
                  ymin = conf.low,
                  ymax = conf.high,
                  fill = ant_species),
              alpha = 0.4,
              # not adopting aesthetic from ggplot
              inherit.aes = FALSE) +
  # group tables based on ant species and relabel
  facet_wrap(~ ant_species,
             labeller = labeller(
               ant_species = c(RRB = "C. mimosae",
                               AB = "C. sjostedti",
                               BBR = "C. nigriceps")
             )) +
  # change colors of graph lines depending on ant species
```

```r
scale_color_manual(values = c("RRB" = "coral",
                              "AB" = "darkolivegreen1",
                              "BBR" = "deeppink2")) +
# change the colors of ribbons
scale_fill_manual(values = c("RRB" = "coral",
                             "AB" = "darkolivegreen1",
                             "BBR" = "deeppink2")) +
# break up lines by 100, not 200
scale_x_continuous(breaks = c(100, 200, 300, 400, 500, 600)) +
# relabel x- and y-axes
labs(x = "Height (cm)",
     y = "Probability of Birds Nests") +
# change theme to `classic`
theme_classic() +
# remove legend
theme(legend.position = "none") +
# remove color guides
guides(
  color = "none",
  fill = "none"
)
```



i. Write a caption for your figure

Figure 1. The higher up in the trees with *C. nigriceps* ant populations, the more likely birds will nest. Points represent the observation of either a nest within the observed tree, or it being available for birds to nest in it and currently without a nest. Each colored line represents the probability of finding a birds nest at a given height (cm). Each colors and graph differentiates the ant population coexisting on the observed trees(green: *C. sjostedti*, orange: *C. mimosae*, pink: *C. nigriceps*), and the faded bands, or ribbons, around the lines represent the 95% confidence interval, or the precision of the predictions. Data extracted from: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. *Zenodo.* https://doi.org/10.5281/zenodo.8373322.

j. Calculate model predictions

```r
# predict occurrence of nests using model 2
ggpredict(nest_mod2,
          terms = c("height_cm [600]", # at 600 cm
                    "ant_species")) # differentiates between ant species
```

```
# Predicted probabilities of case_control

ant_species: AB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.20 | 0.00, 0.99

ant_species: RRB

height_cm | Predicted |     95% CI
----------------------------------
      600 |      0.74 | 0.48, 0.90

ant_species: BBR

height_cm | Predicted |     95% CI
----------------------------------
      600 |      1.00 | 1.00, 1.00
```

k. Interpret your results

Height in the trees (cm) and ant species interact to predict the probability of bird nesting. As height increases, the probability of observing a nest increases. Additionally, the effect of height on nesting probability is stronger for *C. nigriceps* than either *C. mimosae* or *C. sjostedti*.

These relationships can be justified biologically speaking because birds mostly nest in trees with more aggressive ant populations (*C. nigriceps* and *C. mimosae*) to reduce predation of eggs and conceal them in denser cover, and higher up to avoid ground-dwelling predators such as snakes and raccoons.

At 600cm, the probability of nesting is 0.20 (95% CI: [0.00, 0.99]) for trees with *C. sjostedti*, 0.74 (95% CI: [0.48, 0.90]) for trees with *C. mimosae*, and 1.00 (95% CI: [1.00, 1.00]) for trees with *C. nigriceps*.

Problem 3

a. Comparing visualizations

In my homework 2 visualization, I used a bar graph to represent the time I spent biking each day of the week (only one point at this point in time). In my affective visualization, the length of the bike ride was not determined by a bar chart, but instead is shown by a certain scribble design. I did not account of number of per day in my final affective visual, aside from a mention of my typical schedule.

I still used the variables time spent biking (min) and days of the week in my affective visualization, and included the same color scheme of pink and blue. In terms of aesthetics, there are no similarities between the graphs.

Given that I only had one point when making the homework 2 visualization, the provided points were also the means and medians. However, one shift that I generally noticed is that I spent more time biking on campus on Mondays as classes were cancelled and I had to miss class for sickness, even though I had many more classes on Tuesday, Wednesday, and Thursday. So, there was a shift through time in terms of daily biking trends.

I had two ideas for my affective visual – one entailing highlighting the times of day when I was biking the most on a loopy but linear model, and the other which I ultimately decided to pursue, a radial graph that visualized the time that I spent biking as well as the time of day. While I had proposed the first one to Jared and Steven, they both ultimately decided the second idea was better to address my research question. I took their advice and created an affective visual that shows the day I biked, how long I biked for, what time of day I biked, and how these changed over the five weeks of data recording.

b. Designing an analysis

I could use a Kruskal-Wallis test to compare the median amount of time I spend biking per day dependent on the number of classes I have per day.

A Kruskal-Wallis is appropriate, as I am comparing the median biking time spent across four levels that stand for number of classes per day. This would adequately address my question: How does the number of classes I have per day affect the amount of time I spend on average biking?

The variables that I would use in this analysis are time spent biking (min), which is continuous and the response variable, and number of classes per day, which would be categorical and the predictor variable.

c. Check any assumptions and run your analysis

Assumptions for a Kruskal-Wallis test were met, as observations were independent from each other biking throughout the day over many weeks, the independent variable is categorical, and the dependent variable is continuous.

However, it is worth mentioning that unequal sample sizes presented across class categories (1 class has 10 observations, while 4 classes has 1) creates limitations to the accuracy of any data outputs.

```r
# cleaning data to make names shorter and lowercase
bike_clean <- bike_data |>
  rename(biking_time = Time.Spent.Biking..min.,
         classes = Number.of.Classes.Per.Day)
```

```r
# running kruskal-wallis test
kruskal.test(biking_time ~ factor(classes), data = bike_clean)
```

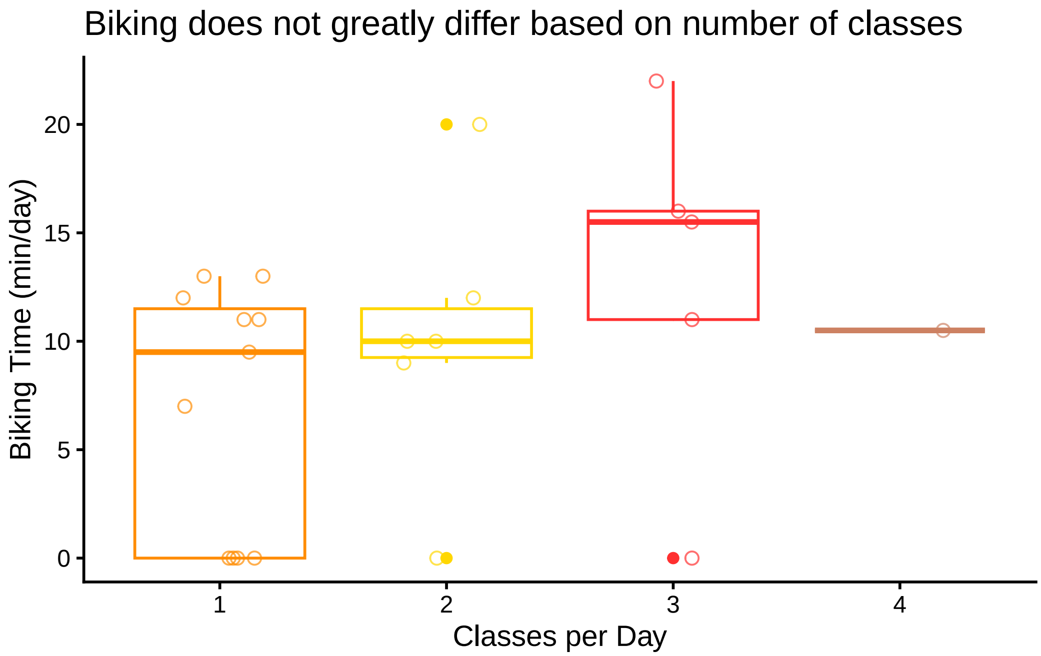
```
	Kruskal-Wallis rank sum test

data:  biking_time by factor(classes)
Kruskal-Wallis chi-squared = 2.9042, df = 3, p-value = 0.4066
```

```r
# determining effect size amongst biking data
rstatix::kruskal_effsize(biking_time ~ factor(classes), data = bike_clean)
```

```
# A tibble: 1 x 5
  .y.             n  effsize method  magnitude
* <chr>       <int>    <dbl> <chr>   <ord>
1 biking_time    23 -0.00504 eta2[H] small
```

d. Create visualization

```
# base layer: ggplot
ggplot(data = bike_clean,
       # x-variable: classes per day
       # y-variable: time spent biking per day
       # change colors dependent on number of classes
       aes(x = factor(classes),
           y = biking_time,
           color = factor(classes))) +
  # first latera: geom boxplot
  geom_boxplot() +
  # second layer: geom jitter to visualize individual data points
  # set height to 0 to avoid vertically spread and y-value changes
  # change size, horizontal spread, transparency, and shape of points
  geom_jitter(height = 0,
              width = 0.2,
              size = 2,
              alpha = 0.7,
              shape = 21) +
  # determine individual colors for classes
  scale_color_manual(values = c("1" = "darkorange",
                                "2" = "gold",
                                "3" = "firebrick1",
                                "4" = "lightsalmon3")) +
  # relabel x-a and y-axes and add title
  labs(x = "Classes per Day",
       y = "Biking Time (min/day)",
       title = "Biking does not greatly differ based on number of classes") +
  # change theme to `classic`
  theme_classic() +
  # remove legend
  theme(legend.position = "none")
```



e. Write caption

Figure 2. The average amount of time spent biking per day does not differ depending on the number of classes attended. The boxplots represent a median amount of minutes spent biking per day given the number of classes over five weeks. This also displays the interquartile range (IQR) and the top and bottom of the box, and outliers of the data shown as opaque points that are not shown in the whiskers. The colors differentiate the number of classes had in a given day (1: orange, 2: yellow, 3: red, 4: orange-brown).

f. Write about your results

We found no significant difference (η^2 = -0.005) in median biking per day taking into account number of classes (Kruskal-Wallis test, $\chi^2(3) = 2.9042$, p = 0.41, $\alpha = 0.05$). From this point, we fail to reject the hypothesis that the number of classes had per day affects the amount of time I spend biking per day.

g. Sharing your affective visualization

I attended workshop 10 and displayed my affective visual with an artist statement.